RESTRICTION ESTIMATES USING MULTILINEAR KAKEYA

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Abstract. We develop a streamlined framework for proving restriction estimates, which yields stronger results than those obtained by polynomial partitioning methods in higher dimensions.

1. Introduction

Suppose $n \geqslant 5$. Let B^{n-1} be the unit ball in $\mathbb{R}^{n-1}$, and define

$$Ef(x', x_n) = \int_{B^{n-1}} e^{i(x'\cdot\xi + x_n|\xi|^2)} f(\xi)\, d\xi.$$

The restriction problem asks for the range of p for which

$$\|Ef\|_{L^p(\mathbb{R}^n)} \leqslant C_{n,p}\|f\|_{L^p(B^{n-1})}. \tag{1.1}$$

The conjectured range is $p > 2n/(n-1)$. Polynomial partitioning has led to a sequence of advances in this problem; see, for example, [5, 7, 4]. Wang–Wu [10] introduced a different approach based on two-ends incidence geometry and refined decoupling. We combine this approach with the broad–narrow method [1, 5], inserting Zahl's multilinear Kakeya estimate [11] directly into the broad argument. This avoids the loss that can occur when the multilinear estimate is first converted into a linear Kakeya maximal estimate.

Fix

$$2 \leqslant k \leqslant \left\lfloor \frac{n+2}{2} \right\rfloor, \qquad d = d_{n,k} := \frac{n^2 + k^2 + n - k}{2n}, \tag{1.2}$$

and put $D = (4n-5)(d-1) + (n-1)(n-3)$. Define

$$p_b(n,k) = 2 + \frac{4(2d+n-5)}{D}, \tag{1.3}$$

$$p_N(n,k) = 2 + \frac{4}{2n-k}, \tag{1.4}$$

Here p_b is the threshold for the transverse broad estimate, whereas p_N is the threshold required by the narrow induction.

Theorem 1.1. *For $n \geqslant 5$, the estimate* (1.1) *holds whenever*

$$p > \min_{2\leqslant k\leqslant \lfloor (n+2)/2\rfloor} \max\{p_b(n,k), p_N(n,k)\} = 2 + \frac{2.58631445\ldots}{n} + O\left(\frac{1}{n^2}\right).$$

For $n = 9$, our result yields the best-known restriction exponents $p > 2150/929 = 2.314316469\ldots$.

The broad–narrow step is a substitute for the higher-dimensional planebrush input that would be needed for a direct geometric approach. Such an estimate could instead strengthen the Wang–Wu framework beyond the present method.

1.1. **Structure of the article.** Section 2 records the required geometric and analytic tools. Section 3 establishes the multilinear incidence estimate and the transverse analytic bound. Section 4 proves the mixed broad estimate and completes the mixed broad-to-linear induction.

1.2. **Notation and conventions.** We write $A \lesssim B$ when $A \leqslant CB$ with a constant depending only on fixed parameters. At a geometric scale δ, the notation $A \lessapprox B$ allows $C_\epsilon \delta^{-\epsilon}$ for every $\epsilon > 0$, together with any explicitly stated $\delta^{-C\epsilon_1}$ loss from a two-ends hypothesis; the analogous convention is used at physical scale r. Fixed powers of dynamic densities or angles are not absorbed in this notation. We keep logarithmic losses explicit when tracking the broad parameter. A term $\mathrm{RapDec}(r)$ is bounded by $C_N r^{-N}$ for every prescribed N.

2. Preliminaries

2.1. **Two-ends shadings.** A unit δ-tube is a fixed constant enlargement of a unit line segment by transverse radius δ. We represent paraboloid tube directions with positive last coordinate. An m-parallel, direction-separated family has at most m tubes in each δ-separated angular class; bounded direction colorings account for buffered caps.

A shading $Y(T)$ is a union of cells of a fixed δ-grid contained in a fixed enlargement of T. Let

$$E_Y = \bigcup_{T\in\mathbb{T}} Y(T), \qquad \mu_Y(x) = \sum_{T\in\mathbb{T}} \mathbb{1}_{Y(T)}(x).$$

The shading is λ-dense if $|Y(T)| \sim \lambda|T|$ for every retained tube. Later, for a refined subfamily, λ may instead denote the explicitly defined *average* density; this does not impose a new two-ends hypothesis on that subfamily.

Definition 2.1 (Two ends)**.** *For $0 < \epsilon_2 < \epsilon_1 < 1$, a shading is (ϵ_1, ϵ_2)-two-ends if*

$$|Y(T) \cap B(x, \delta^{\epsilon_1})| \leqslant C\delta^{\epsilon_2}|Y(T)|$$

for every T and every such ball. Harmless fixed dilations and fixed constants are incorporated by decreasing ϵ_2 slightly. The constants in the geometric inputs may depend on ϵ_2.

The Wolff axiom with error m means that, for every box of dimensions $1 \times s_1 \times \cdots \times s_{n-1}$, with $\delta \leqslant s_j \leqslant 1$, the number of tubes contained in a fixed dilation of the box is at most

$$Cm\, s_1 \cdots s_{n-1} \delta^{-(n-1)}. \tag{2.1}$$

An m-parallel direction-separated family has this property: a contained tube must point in an angular region having the indicated product size, and each allowable direction contributes at most m tubes. In particular,

$$\delta^{n-1}\#\mathbb{T} \lesssim m.$$

2.2. **Geometric tools.** We use the following two geometric tools with their small-loss versions.

Theorem 2.2 (General two-ends hairbrush; Gao–Wu–Xi)**.** *Suppose that $\mathbb{T}$ satisfies* (2.1) *and has a λ-dense, (ϵ_1, ϵ_2)-two-ends shading. Then* [3, Lemma 4.1] *gives*

$$|E_Y| \gtrsim_\epsilon \delta^{\epsilon + C\epsilon_1} m^{-\frac{n}{2n-2}} \lambda^{7/4} \delta^{(n-2)/2} (\delta^{n-1}\#\mathbb{T})^{\frac{n}{2n-2}}.$$

There is a good set $E_H \subset E_Y$ with $|E_Y \backslash E_H| \leqslant \delta^{\epsilon_1}|E_Y|$ on which

$$\mu_Y \lesssim_\epsilon \delta^{-\epsilon - C\epsilon_1} m^{\frac{n}{2n-2}} \lambda^{-3/4} \delta^{-(n-2)/2} (\delta^{n-1}\#\mathbb{T})^{\frac{n-2}{2n-2}}.$$

Consequently, for an m-parallel direction-separated family,

$$\mu_Y \lessapprox m\lambda^{-3/4}\delta^{-(n-2)/2} \quad \text{on } E_H. \tag{2.2}$$

Only this general form of the hairbrush estimate is used below.

Theorem 2.3 (Direction-separated multilinear Kakeya; Zahl)**.** *Let $2 \leqslant k \leqslant n$ and $d = d_{n,k}$. For a direction-separated family of unit δ-tubes,* [11, Theorem 1.3] *gives*

(2.3)

$$\left\| \sum_{T_1,\dots,T_k \in \mathbb{T}} \mathbb{1}_{T_1} \cdots \mathbb{1}_{T_k} |v(T_1) \wedge \cdots \wedge v(T_k)|^{k/d} \right\|^{1/k}_{d/k(d-1)} \lesssim_\epsilon \delta^{1-n/d-\epsilon} (\delta^{n-1}\#\mathbb{T})^{\frac{n(d-1)}{(n-1)d}}.$$

We use (2.3) directly, without invoking the multilinear-to-linear Kakeya reduction.

2.3. **Broad norms.** Fix $R \gg 1$ and a dyadic $K \asymp R^\kappa$, with $\kappa > 0$ small. Let τ range over a finitely overlapping cover of B^{n-1} by K^{-1}-caps. We write $\boldsymbol{f} = (f_\tau)_\tau$, where $\operatorname{supp} f_\tau \subset 2\tau$, and use the notation

$$\|\boldsymbol{f}\|_2^2 = \sum_\tau \|f_\tau\|_2^2, \qquad \|\boldsymbol{f}\|^2_{L^2_{\text{avg}}(\theta)} = \frac{1}{|\theta|}\sum_\tau \int_\theta |f_\tau|^2. \tag{2.4}$$

For a smooth cap decomposition of f, these quantities are bounded by the corresponding quantities for f, up to a fixed constant.

For $r \geqslant 1$, write

$$M_r(f) := \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\text{avg}}(\theta)}.$$

We use the same notation for $\boldsymbol{f}$, with local averages interpreted as in (2.4).

For a cube b of side K^2, define

$$\mu_{k,A,p}(E\boldsymbol{f}; b) = \inf_{V_1,\dots,V_A} \max_{\substack{\tau:\, \not\sphericalangle(v(2\tau),V_a)>K^{-1} \\ \text{for all } 1\leqslant a\leqslant A}} \int_b |Ef_\tau|^p, \tag{2.5}$$

where the infimum is over $(k-1)$-dimensional linear subspaces of $\mathbb{R}^n$, and $v(2\tau)$ denotes the tube directions associated with 2τ. The maximum is zero if there is no eligible cap. Set

$$\|E\boldsymbol{f}\|^p_{\mathrm{BL}^p_{k,A,K}(U)} = \sum_{b\subset U} \mu_{k,A,p}(E\boldsymbol{f}; b).$$

We use a fixed grid and take larger cubes to be unions of its K^2-cubes. The same definition applies when the functions Ef_τ are replaced by other functions indexed by the same caps.

At an intermediate scale r, wave-packet decompositions enlarge frequency supports by $O(r^{-1/2})$. We keep the original cap indices and choose the scales so that these supports remain in 2τ.

The broad norm satisfies the triangle inequality with a loss in the parameter A; see [5, (1.15)].

Lemma 2.4 (Broad triangle inequality)**.** *Let* $\boldsymbol{f} = \sum_{j=1}^{L} \boldsymbol{f}_j$ *and* $A_1 + \cdots + A_L \leqslant A$. *Then*

$$\|E\boldsymbol{f}\|_{\mathrm{BL}^p_{k,A,K}(U)} \leqslant \sum_{j=1}^{L} \|E\boldsymbol{f}_j\|_{\mathrm{BL}^p_{k,A_j,K}(U)}.$$

By the standard reduction from k-broad estimates to multilinear estimates (see [5, Section 1.2]), the broad contribution is bounded, up to $K^{O(1)}$, by a sum over $K^{O(1)}$ tuples of quantitatively transverse caps. For each such tuple, writing Ef_i for the corresponding contributions. Then the Minkowski inequality gives

$$\sum_{b\subset X} \min_{1\leqslant i\leqslant k} \int_b |Ef_i|^p \lesssim \left(\prod_{i=1}^{k} \int_X |Ef_i|^p\right)^{1/k}. \tag{2.6}$$

When partitioning X by multiplicity, we first retain the sum on the left of (2.6), which is additive over the cubes, and apply Hölder only after choosing a piece. For wave-packet sums, the smaller caps are formed by grouping packets according to their directions. We require $r^{-1/2} \ll K^{-L}$ so that their frequency supports remain transverse.

2.4. **Wave packets, local averages of the L^2-norm, and refined decoupling.** At scale r, decompose each f_τ into wave packets f_T associated with $r^{-1/2}$-caps θ and tubes T of length r and radius $r^{1/2+\epsilon_0}$. The cap indices are retained throughout. We suppress rapidly decreasing errors and keep the losses $r^{O(\epsilon_0)}$ caused by the tube enlargement.

After a bounded decomposition of the frequency caps and spatial lattice, we may assume two-sided almost orthogonality: for every packet subfamily $\mathcal{S}$,

$$\left\|\sum_{T\in\mathcal{S}} f_T\right\|_2^2 \sim \sum_{T\in\mathcal{S}} \|f_T\|_2^2, \qquad \sum_{T} \|f_T\|_2^2 \lesssim \|\boldsymbol{f}\|_2^2.$$

Together with frequency localization, this gives, for $1 \leqslant t \leqslant r$,

$$M_t\left(\sum_{T\in\mathcal{S}} f_T\right) \lesssim M_r(\boldsymbol{f}). \tag{2.7}$$

These estimates apply cap by cap, with the norms interpreted as in (2.4) when the cap indices are retained.

After dyadic pigeonholing, assume

$$\|f_T\|_2 \asymp a, \qquad \#\{T : \theta(T) = \theta\} \asymp m \tag{2.8}$$

for each occupied cap. Then

$$a^2 m \lesssim r^{-(n-1)/2} M_r(\boldsymbol{f})^2, \qquad a^2 \#\mathbb{T} \lesssim \|\boldsymbol{f}\|_2^2. \tag{2.9}$$

Further subfamilies have at most $O(m)$ packets in each cap and still satisfy (2.9). The pigeonholing costs at most $(\log r)^{O(1)}$, including the loss in A from Lemma 2.4.

Let X be a union of $r^{1/2+\epsilon_0}$-cubes, and let

$$f_i = \sum_{T\in\mathbb{T}_i} f_T, \qquad N_i = \#\mathbb{T}_i,$$

where $\mathbb{T}_i$ is a fixed packet subfamily. For each $T \in \mathbb{T}_i$, let $Y_i(T)$ be the union of the cubes of X meeting T, and define

$$\sum_{T\in\mathbb{T}_i} |Y_i(T)| = \lambda_i \sum_{T\in\mathbb{T}_i} |T|.$$

Fixed dilations of cubes and tubes are understood. Local L^2 orthogonality on each cube, summed over X, yields

$$\int_X |Ef_i|^2 \lesssim r^{O(\epsilon_0)}\lambda_i r \, \|f_i\|_2^2. \tag{2.10}$$

Set $q = 2(n+1)/(n-1)$. If every cube of X meets at most μ_i tubes from $\mathbb{T}_i$, refined decoupling [6, Theorem 4.2], in the form [10, Theorem 4.4], gives

$$\begin{aligned}\int_X |Ef_i|^q &\lesssim_\epsilon r^{\epsilon+O(\epsilon_0)}\mu_i^{2/(n-1)} a^q N_i \\ &\lesssim r^{-1+\epsilon+O(\epsilon_0)}(\mu_i/m)^{2/(n-1)}\|f_i\|_2^2 M_r(\boldsymbol{f})^{q-2}.\end{aligned} \tag{2.11}$$

Here $\|Ef_T\|^q_{L^q(w_{B_r})} \lesssim a^q$ follows by interpolating the standard packet bounds $\|Ef_T\|_\infty \lesssim r^{-(n-1)/4}a$ and $\|Ef_T\|^2_{L^2(w_{B_r})} \lesssim ra^2$. The second inequality follows from (2.9). Any dyadic summation over packet L^q-norms is absorbed into r^ϵ.

3. Multilinear two-ends estimates

Fix n, k as in (1.2). Throughout this section,

$$d = d_{n,k}, \qquad q = \frac{2(n+1)}{n-1}, \qquad D = (4n-5)(d-1) + (n-1)(n-3),$$

and p_b, p_N are defined in (1.3)–(1.4). We first estimate the multiplicity of transverse tube families without assuming that their populations are comparable, then combine this estimate with the hairbrush bound and refined decoupling.

3.1. **A geometric lemma allowing unequal transverse populations.**

Lemma 3.1. *Let $\mathbb{T}_1, \dots, \mathbb{T}_k$ be disjoint angular families of unit δ-tubes, and suppose every choice $T_i \in \mathbb{T}_i$ has wedge at least $\nu > 0$. There are at most m tubes in each δ-separated direction class. Let E be a union of $O(\delta^{-n})$ grid cells of side δ. Assume the shaded multiplicity in family i is comparable to a positive constant μ_i on every cell of E. Set*

$$N_i = \#\mathbb{T}_i, \qquad N = \sum_i N_i, \qquad Z = \frac{\delta^{n-1}N}{m},$$

$$\zeta = \frac{(N_1\cdots N_k)^{1/k}}{N}, \qquad \Lambda = (\lambda_1\cdots\lambda_k)^{1/k}, \qquad \mu = (\mu_1\cdots\mu_k)^{1/k}.$$

where $\mu_i|E| \asymp \lambda_i\delta^{n-1}N_i$ defines the average densities. If $\min_i \mu_i/m \geqslant C_{n,k}\log(2/\delta)$, then

$$\frac{\mu}{m} \lesssim_\epsilon \nu^{-1}\delta^{-(n-d)-\epsilon}\Lambda^{-(d-1)}\zeta^{-(d-1)}Z^{(d-1)/(n-1)}. \tag{3.1}$$

Proof. Pad each direction class with dummy entries to have m slots, and choose one slot uniformly and independently in each class. For each fixed cell and family i, the sampled shaded multiplicity is a sum of independent Bernoulli variables, one for each direction class, of total expectation comparable to μ_i/m. Chernoff's inequality and a union bound over the polynomially many cells show that all these

sampled multiplicities are at least $c\mu_i/m$ simultaneously, with probability greater than 3/4, when the displayed logarithmic threshold is large enough. The expected total number of sampled tubes is N/m; Markov's inequality makes it at most $8N/m$ with probability at least 7/8. Thus a realization with both properties exists. A bounded direction coloring handles adjacent direction cells when necessary.

The chosen realization is direction-separated. Apply (2.3) to its complete tubes, using their shaded incidences only for a lower bound on the integrand on E. Raise the result to $d/(d-1)$:

$$\nu^{1/(d-1)}(\mu/m)^{d/(d-1)}|E| \lesssim_\epsilon \delta^{-(n-d)/(d-1)-\epsilon} Z^{n/(n-1)}. \tag{3.2}$$

Taking geometric means of the incidence identities gives

$$\mu|E| \sim \Lambda\zeta\delta^{n-1}N = \Lambda\zeta mZ.$$

Substitute this into (3.2), cancel, and raise to the power $d-1$. This proves (3.1), after renaming the arbitrary small loss. □

4. Proof of the main theorem

4.1. **The broad two-ends induction.** We prove the broad estimate by a two-ends induction for the broad measures (2.5), combining Lemma 3.1 with the hairbrush bound and refined decoupling in the two-ends case. The outer scale remains fixed while the intermediate radius decreases, and a slowly growing broad parameter pays for the dyadic decompositions; compare [2, Proposition 4.13].

Proposition 4.1. *Fix n, k as in Theorem 1.1. Given $\eta > 0$, one can choose $\kappa > 0$ sufficiently small and then $a_0 > 0$ arbitrarily small such that, with $K \asymp R^\kappa$, the following holds:*

$$\|E\boldsymbol{f}\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_R)} \lesssim_\eta R^{2\eta}\|\boldsymbol{f}\|_2^2 M_R(\boldsymbol{f})^{p_b-2}, \qquad A \geqslant R^{a_0}. \tag{4.1}$$

All constants are independent of R and A. For canonical sector decompositions, $\|\boldsymbol{f}\|_2 \lesssim \|f\|_2$.

Proof. **Parameter order.** Choose a stopping exponent $\theta > 0$ with $n\theta < \eta/4$. Choose a time-shortening exponent $\alpha > 0$ small enough that every geometric loss $r^{C_{n,k}\alpha}$ below is bounded by $r^{\eta/4}$. Choose $b_0 > 0$ with $2b_0 < \eta\alpha/4$. Take the wave-packet enlargement exponent and the arbitrary losses in (2.3) and refined decoupling much smaller than b_0 and η. Choose κ so small that the extraction loss $K^{C_{n,k}} \leqslant R^{\eta/4}$ and $K^{2L} \ll R^\theta$. Finally take $a_0 > 0$ as small as desired, but fixed and positive. Increase the lower threshold for R so that every polylogarithmic loss is dominated by the positive powers used below.

For $r \in [R^\theta, R]$ we prove

$$\|E\boldsymbol{f}\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)} \leqslant C_\eta R^\eta r^\eta \|\boldsymbol{f}\|_2^2 M_r(\boldsymbol{f})^{p_b-2}, \qquad A \geqslant r^{a_0}. \tag{4.2}$$

Use a discrete sequence with successive radii r and $r_1 = r^{1-\alpha}$, rounded dyadically. Its length down to R^θ is bounded in terms of η alone. For an intermediate requested radius, start the descending sequence at that radius.

Base case. For $r \lesssim R^\theta$, the trivial bound is $Cr^n\|\boldsymbol{f}\|_2^{p_b}$. Since $\|\boldsymbol{f}\|_2 \lesssim M_r(\boldsymbol{f})$ and $r^n \leqslant CR^{\eta/4}$, this is absorbed by (4.2).

Pigeonholing. Select a union X of $r^{1/2+\epsilon_0}$-cubes on which the broad mass per cube is comparable and which carries the original broad norm up to logarithms.

Partition the x_n axis into intervals of length $r_1 = r^{1-\alpha}$, aligned with the physical grid. For every tube, define its segments using these *same* time intervals. Select a dyadic λ for which each retained segment J meets X in volume comparable to $\lambda|T|$. Select a dyadic number β of such segments per retained tube. Necessarily $1 \leqslant \beta \lesssim r^\alpha$. Let

$$Y(T) = \bigcup_{J \text{ selected for } T} (J \cap X \cap T)$$

with grid-cell shadings in a fixed tube enlargement. Its density is comparable to $\lambda_0 = \beta\lambda$. On a fixed common time interval the truncated sum is exactly the extension of the subfamily whose interval is active, modulo the already controlled spatial tails. There is no pointwise monotonicity assertion when deleting packets.

The selected broad measure is carried by a set $X_2 \subset X$, with

$$|X_2| \gtrsim (\log r)^{-C}|X|, \tag{4.3}$$

and with comparable selected broad mass on its constituent cubes. The broad parameter after these selections is at least $A/(\log r)^C$.

Case 1: The non-two-ends scenario. Suppose $\beta \leqslant r^{b_0}$. Partition each common time interval spatially into cubes of side comparable to r_1. For such a cube B_j, let $\boldsymbol{f}_j$ be the vector of active packets whose tubes meet a fixed enlargement of B_j, retaining sector labels. On B_j the truncated sum agrees with $E\boldsymbol{f}_j$ up to rapidly decreasing errors. A tube meets only $O_n(1)$ spatial cubes in each active interval, since its slope is bounded and its width is $o(r_1)$. Consequently,

$$\sum_j \|\boldsymbol{f}_j\|_2^2 \lesssim r^{b_0+C\epsilon_0}\|\boldsymbol{f}\|_2^2, \qquad M_{r_1}(\boldsymbol{f}_j) \lesssim M_r(\boldsymbol{f}).$$

The average estimate follows from (2.7); the new support buffer absorbs the packet-projection enlargement.

Since

$$\frac{r^{a_0}}{(\log r)^C} \geqslant r^{(1-\alpha)a_0} = r_1^{a_0}$$

for sufficiently large R, the induction hypothesis applies on every B_j. After summation the ratio to the right side of (4.2) is bounded by

$$C(\log r)^C r^{-\eta\alpha+b_0+C\epsilon_0}.$$

The parameter choices make this smaller than $1/2$. This closes the first case.

Case 2: The two-ends scenario. Suppose $\beta > r^{b_0}$. A ball of radius comparable to r_1 meets only $O_n(1)$ of the common time intervals. Since all selected segments have comparable shading size, its share of any $Y(T)$ is at most C/β. Hence the normalized shading is a two-ends shading with parameters comparable to 2α and b_0 at tube scale $\delta \sim r^{-1/2}$, with the small enlargement adjustments already budgeted.

Apply (2.2) to this *ancestor* shading before selecting transverse caps. This gives a union of good grid cubes with exceptional volume at most $r^{-c\alpha}|X|$ and the multiplicity bound (4.4). By (4.3) and the uniform broad mass on X_2, the exceptional set removes less than half of the selected broad mass for sufficiently large R. Restrict to the good set. No further two-ends property is required of subsequent restrictions.

Fix a transverse tuple at cost K^C, then one spatial cube in one common time interval at cost at most $r^{C_n\alpha}$. On that cube the active packet families are fixed complete extensions. Pigeonhole their separate positive multiplicities on the $r^{1/2+\epsilon_0}$ grid while retaining the additive functional $\sum_{b\subset X} \min_i \int_b |Ef_i|^{p_b}$ from (2.6); there

are only polylogarithmically many joint bins. Choose a bin carrying a polylogarithmic fraction of that functional. On that bin, and not before it is chosen, bound the functional by $\mathcal{I}_{p_b}$. Cubes where a family has zero multiplicity contribute only controlled tails to the local minimum functional and are removed. Denote the surviving region again by X, and write $f_1, \dots, f_k$ for the corresponding packet sums, each formed from a fixed subfamily in the selected common time interval, as in Subsection 2.4.

The inherited good-set hairbrush bound for the *ancestor shading* gives

$$\mu_i \leqslant \mu_{\text{ancestor}} \lesssim r^{e} m \lambda_0^{-3/4} r^{(n-2)/4}, \tag{4.4}$$

where $e > 0$ is arbitrarily small apart from a fixed dimensional multiple of the two-ends scale parameter. Since each refined shading is a subset of its ancestor shading, its average density is at most $C\lambda_0$. In particular $\Lambda \lesssim \lambda_0$, and (4.4) remains valid with $\lambda_0^{-3/4}$ replaced by $C\Lambda^{-3/4}$.

We first consider the case in which $\mu_i/m \lesssim r^{C\epsilon_0} \log r$ for some i. By (2.10) and (2.11), Hölder's inequality gives

$$\int_X |Ef_i|^{p_b} \leqslant \left(\int_X |Ef_i|^2 \right)^{1-t} \left(\int_X |Ef_i|^q \right)^{t} \lesssim r^{1-2t+O(\epsilon_0)+\epsilon} \|f_i\|_2^2 M_r(\boldsymbol{f})^{p_b-2},$$

where $t = (p_b - 2)/(q-2) = 1 - u$. Since $1 - 2t < 0$, this gives the required bound for the selected additive functional, using

$$\sum_{b \subset X} \min_j \int_b |Ef_j|^{p_b} \lesssim \int_X |Ef_i|^{p_b}.$$

In the remaining case, all the ratios μ_i/m exceed this threshold, so Lemma 3.1 applies. Put

$$\mathcal{I}_q := \left(\prod_{i=1}^{k} \int_X |Ef_i|^q \right)^{1/k}.$$

Taking geometric means in (2.10) and combining (2.11) with the hairbrush estimate (4.4) and Lemma 3.1, respectively, we obtain

$$\begin{aligned} &\mathcal{I}_2 \lesssim r^{e+1} \Lambda \|\boldsymbol{f}\|_2^2, \qquad \mathcal{I}_q \lesssim r^{-n/(2(n-1))+e} \Lambda^{-3/(2(n-1))} \|\boldsymbol{f}\|_2^2 M_r(\boldsymbol{f})^{q-2}, \\ &\mathcal{I}_q \lesssim K^C r^{-(d-1)/(n-1)+e} \Lambda^{-2(d-1)/(n-1)} \|\boldsymbol{f}\|_2^2 M_r(\boldsymbol{f})^{q-2}, \end{aligned} \tag{4.5}$$

since $Z, \zeta \lesssim 1$. The polynomial K loss includes the wedge bound in (3.1).

Choose

$$u = \frac{(2n-3)(d-1)}{D}, \qquad v = \frac{2(n-1)(d-1)}{D}, \qquad w = \frac{(n-1)(n-3)}{D}.$$

By Hölder and (4.5),

$$\mathcal{I}_{p_b} \lesssim \mathcal{I}_2^u I_q^v I_q^w \lesssim K^C r^e \|\boldsymbol{f}\|_2^2 M_r(\boldsymbol{f})^{p_b-2}.$$

In either case, restoring the losses from the preceding selections yields

$$\|E\boldsymbol{f}\|_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)}^{p_b} \lesssim_\eta K^{C_{n,k}} r^{C_{n,k}\alpha + o_{\epsilon_0}(1)+\epsilon'} (\log r)^C \|\boldsymbol{f}\|_2^2 M_r(\boldsymbol{f})^{p_b-2},$$

where the arbitrary input loss $\epsilon' > 0$ was chosen sufficiently small. The parameter choices absorb this into $C_\eta R^\eta r^\eta \|\boldsymbol{f}\|_2^2 M_r(\boldsymbol{f})^{p_b-2}$. The constant C_η can be enlarged once to cover the base scales. This completes the induction and proves (4.1). $\square$

4.2. **From mixed broad estimates to linear restriction.** Fix

$$\max\{p_b, p_N\} < p < \frac{2(k-1)}{k-2}.$$

By Proposition 4.1,

$$\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} \lesssim_{\eta,p} R^{2\eta}\|f\|_2^2\|f\|_\infty^{p-2}, \qquad K \sim R^\kappa, \quad A \geqslant R^{a_0}.$$

The broad–narrow decomposition and [5, Lemma 9.3] give, for every $\delta_N > 0$,

$$\int_{B_R} |Ef|^p \lesssim K^{C_{n,p}}\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} + C_{\delta_N} A^{p-1} K^{\delta_N + (k-2)(p/2-1)} \sum_\tau \int_{B_{2R}} |Ef_\tau|^p.$$

Since a_0 may be chosen arbitrarily small after fixing κ, the parabolic rescaling argument in [5, Section 9] applies. The proof is then completed by restricted-type real interpolation and a standard ϵ-removal argument (see [9]).

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